The Dirichlet Character Atlas

— A Weil-Kernel Numerical Atlas for Dirichlet L-Function Sector Gaps

Galerkin Diagnostics and the Boundaries of Leading-Order Theory

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Abstract

This paper is an instantiation of Functional Stability Theory (FST) in the mathematical domain: it extends the three-component decomposition $\mathrm{RH}(X) = \mathrm{GBC}(X) + C^{(2)}(X) + \mathrm{Hurwitz}(X)$ (see Math-Master [1], §2.4) to twisted L-functions, where the sector gap is the quantitative signature of $C^{(2)}_\chi$. The Math-Master [1] develops the overall zeta-zoo architecture and the trinity UCU + SGE + Weil-form transversality; the Physics-Master [2] develops the companion physical formulation via the Renormalized Free Energy Principle.

We develop a Weil-kernel numerical atlas of sector gaps for Dirichlet L-functions associated to ten low-conductor real characters χ_D, $D \in \{5, 8, 12, 13, 17, 21, 24, 29, 33, 60\}$, at Galerkin truncations $N \in \{200, 400, 600\}$ and cut-off $\lambda = 20000$. Within the three-component decomposition the sector gap is the quantitative signature of $C^{(2)}_\chi$. Our main results are *negative but structurally informative.* First, the leading-order approximation $\phi^+_\chi \approx \phi^-_\chi$ of the two sector ground states is empirically falsified: a systematic 16-window test, including the empirical ground state itself, fails the success criterion 9/10 sign accuracy, $R^2 \geq 0.8$. Second, the ostensibly "exact" formula

$$\mathrm{gap}_\chi = -2\sum_{p,m} \frac{\chi(p)^m \log p}{p^{m/2}} \Delta_\chi(m \log p) + \mathrm{arch\_diff}_\chi + O(N^{-1})$$

with $\Delta_\chi(t) := (\phi^+_\chi * \phi^+_\chi)(t) + (\phi^-_\chi * \phi^-_\chi)(t)$ reduces, once both ground states are taken from the same Galerkin diagonalization, to a matrix-decomposition tautology: an apparent 9/9 sign accuracy at $N = 200$ disappears as a numerical interpolation artifact when the analysis is repeated at $N = 600$. Third, the χ_{21} gap undergoes a sign flip between $N = 200$ and $N = 600$ ($+0.282 \to -0.004$), illustrating that even the Galerkin operator itself is not informative below its convergence scale. We catalog the ten characters along two filter constants (parity and root number, both $+1$ in the sample) and two informative axes (archimedean factor weight and gap signature), exhibit the systematic Siegel-Walfisz compression at $N = 600$, and position the result as a precise diagnostic of why numerical Galerkin routes cannot, by themselves, resolve $C^{(2)}_\chi$. A genuine predictive formulation would require Paley-Wiener asymptotic control of the sector ground states that bypasses the Galerkin truncation; we close with a roadmap toward such a branch-local Hilbert-Pólya framework.

I Introduction

A Motivation

The current even-dominance reduction programme for the Riemann hypothesis [4] represents the even-parity sector of the Weil quadratic form on $L^2[-L, L]$, $L = \log\lambda$, as the route to dominance over the odd-parity sector uniformly in λ, at the target rate $\mathrm{gap}(\lambda) \gtrsim \sqrt{\lambda}$. The construction is presented there as a conditional reduction and diagnostic route, not as a closed unconditional proof; it is variational and uses only the three structural ingredients of $\zeta(s)$: the Euler product, the Gamma factor, and a single functional equation.

A natural and central question is whether the same architecture transfers to twisted L-functions $L(s, \chi)$ for non-trivial Dirichlet characters χ, thereby addressing the Generalized Riemann Hypothesis (GRH) one character at a time. In the meta-programme of [5], the RH-type question for a zeta family X is diagnosed through a three-component decomposition

$$\mathrm{RH}_X \;=\; \mathrm{GBC}_X \;+\; C_X^{(2)} \;+\; \mathrm{Hurwitz}_X, \qquad (1)$$

where GBC_X captures the Galerkin bridge to a self-adjoint operator, $\mathrm{Hurwitz}_X$ the reality-of-zeros transfer, and $C_X^{(2)}$ the Connes-style operator identification at the heart of the Hilbert-Pólya programme. For the trivial character, [4] isolates the corresponding even-dominance route for $C^{(2)}_{X=\zeta}$; its closure is treated there as a conditional reduction rather than an unconditional theorem. For non-trivial χ, $C_\chi^{(2)}$ is the hard residual component. The present paper is an atlas of what Galerkin-numerical techniques can and, more importantly, cannot say about this residual.

B What this paper does

We catalogue the Galerkin sector gap $\mathrm{gap}_\chi(\lambda) := \lambda_{\min}(Q_\chi^-) - \lambda_{\min}(Q_\chi^+)$ at $\lambda = 20000$ for ten Kronecker-real characters χ_D, $D \in \{5, 8, 12, 13, 17, 21, 24, 29, 33, 60\}$, across three Galerkin resolutions $N \in \{200, 400, 600\}$. For each character we compute both sector ground states separately $\phi_\chi^\pm$, the primitive Weil kernel prediction

$$S_\chi(\lambda) = -2 \sum_{\substack{p \nmid q \\ m \log p \le 2L}} \frac{\chi(p)^m \log p}{p^{m/2}}\, \Delta_\chi(m \log p), \qquad (2)$$

with $\Delta_\chi(t) = (\phi_\chi^+ * \phi_\chi^+)(t) + (\phi_\chi^- * \phi_\chi^-)(t)$, the archimedean contribution $\mathrm{arch\_diff}_\chi := \langle \phi^-, K_-^{\mathrm{arch}} \phi^- \rangle - \langle \phi^+, K_+^{\mathrm{arch}} \phi^+ \rangle$, and the full Galerkin gap $\mathrm{gap}_{\mathrm{gal}}^{(N)}$. We position the four quantities relative to each other, across all three Galerkin resolutions, and relative to the empirical (highest-N) gaps as a reference.

C What this paper finds

Our contribution is deliberately diagnostic. The key findings are:

(F1) **Falsification of the leading-order reduction.** Under the approximation $\phi_\chi^+ \approx \phi_\chi^- \approx \phi_\infty$ used in [4] for the trivial character, the prime-side predictor reduces to a universal autocorrelation integral. A systematic test of 16 window models (including the empirical single-sector ground state) achieves at best 6/10 sign accuracy and $R^2 = 0.15$. The leading-order reduction is false for non-trivial χ.

(F2) **Matrix-decomposition tautology.** If *both* sector ground states are taken from the same Galerkin diagonalization, the identity $\mathrm{gap}_{\mathrm{gal}}^{(N)} = S_\chi + \mathrm{arch\_diff}_\chi$ is not a prediction but a consequence of the linearity of expectation on $W = W_{\mathrm{arch}} + W_{\mathrm{prime}}$. An apparent 9/9 sign accuracy at $N = 200$ (modulo the known N-oscillator χ_{21}) dissolves under $N = 600$ server-level validation: the predictor $S_\chi + \mathrm{arch\_diff}_\chi$ drops to 8/10 owing to two sign failures: χ_{21} (N-oscillator, see (F3) and §VII) and χ_{29} (numerical interpolation artifact at a near-zero gap).

(F3) **χ_{21} as an N-oscillator.** At $N = 200$ one has $\mathrm{gap}_{\mathrm{gal}} = +0.282$; at $N = 600$ one has $\mathrm{gap}_{\mathrm{gal}} = -0.004$. The sign flip is not a formula defect but a Galerkin truncation effect for composite conductors that only settles above $N \sim 500$. This both demonstrates that the Galerkin operator is itself non-trivially N-dependent, and explains why a 9/9 at $N = 200$ was not surprising once χ_{21} was excluded: the excluded character was the very one whose N-dependence was not yet integrated into the test.

(F4) **Residual content: the Weil-kernel architecture is rigorous.** The sector-kernel decomposition (Theorem II.1), the anti-diagonal sector difference (Theorem II.2), and the spectral-domain form of the gap via the L'/L function remain valid as structural statements independent of the leading-order reduction. We exhibit them in Section II and use them to classify the ten characters along four orthogonal axes in Section V.

Taken together, these findings deliver a precise diagnosis: the $C_\chi^{(2)}$-component of (1), for non-trivial χ, is not accessible to Galerkin-numerical closure because the relevant signal lives in the difference $\phi_\chi^+ - \phi_\chi^-$, which is fine structure lost in every natural leading-order approximation, and because the full Galerkin operator is a tautology relative to its own eigenvectors. A genuinely predictive formulation must identify Δ_χ analytically, e.g. via Paley-Wiener asymptotics [15] adapted from [4] to the character-twisted setting.

D Organization

Section II summarizes the Weil quadratic form and the sector-kernel decomposition. Section III states the gap formula and contrasts the reduced and unreduced versions. Section IV presents the numerical validation at

$N \in \{200, 400, 600\}$. Section V catalogs the ten characters along four axes. Section VI formulates and proves the tautology. Section VII treats the χ_{21} N-oscillation in detail. Section VIII discusses implications for (1). Section IX outlines the Paley-Wiener roadmap and the branch-local Hilbert-Pólya program as the indicated next step.

II Setup: Weil quadratic form and sector kernels

We summarize the structural framework used throughout; the full derivations are given in [6].

A Dirichlet L-functions and the Weil formula

Let χ be a primitive Dirichlet character mod q with $\chi(-1) = +1$ (even). Set $a_\chi = 0$ and define the completed L-function

$$\Lambda(s, \chi) = \Big(\frac{q}{\pi}\Big)^{s/2} \Gamma\Big(\frac{s}{2}\Big) L(s, \chi),$$

satisfying the functional equation $\Lambda(s, \chi) = \epsilon_\chi \Lambda(1 - s, \bar{\chi})$ with $|\epsilon_\chi| = 1$. For a real (i.e., Kronecker) character $\bar{\chi} = \chi$ and $\epsilon_\chi = +1$; we work exclusively with the ten real characters χ_D with $D \in \{5, 8, 12, 13, 17, 21, 24, 29, 33, 60\}$ in this atlas. The non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ are assumed on the critical line throughout (GRH for the tested conductors). The database pages of [16] provide computed low-lying zeros on the critical line for the corresponding Dirichlet L-functions. We use these entries only as finite numerical consistency data; they are not a certificate that all zeros below a specified height have been verified on the critical line. A theoretical proof of GRH for $L(s, \chi_D)$ for any specific D in the sample is, of course, open.

For an even compactly supported test function $h \in C_c^\infty(\mathbb{R})$, the Weil explicit formula [11, 12] reads

$$\sum_\gamma h(\gamma) = W_\infty^\chi[h] - 2 \sum_{\substack{p \nmid q \\ m \geq 1}} \frac{\chi(p)^m \log p}{p^{m/2}} \hat{h}(m \log p), \quad (3)$$

where the sum on the left runs over all imaginary parts of non-trivial zeros of $L(s, \chi)$, and the archimedean weight is

$$W_\infty^\chi[h] = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(t) \Big[\log \frac{q}{\pi} + \Re\, \psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big)\Big]\, dt.$$

B Quadratic form on H_L

Fix $\lambda > 1$ and set $L := \log \lambda$, $H_L := L^2[-L, L]$. For a real test function $f \in C_c^\infty([-L, L])$, put $h(t) := 2\pi |\hat{f}(t)|^2$. Then h is non-negative even, and (3) induces the *Weil quadratic form*

$$\begin{aligned} Q_\chi[f] &:= \sum_\gamma 2\pi |\hat{f}(\gamma)|^2 \\ &= W_\infty^\chi[2\pi |\hat{f}|^2] \\ &\quad - 2 \sum_{p,m} \frac{\chi(p)^m \log p}{p^{m/2}} \int f(y) f(y - m \log p)\, dy. \end{aligned} \quad (4)$$

Q_χ admits a translation-invariant symbol: defining the primitive convolution distribution

$$\begin{aligned} \kappa_\chi(u) := - \sum_{p \nmid q,\, m \geq 1} \frac{\chi(p)^m \log p}{p^{m/2}} \\ \times \big[\delta(u - m \log p) + \delta(u + m \log p)\big], \end{aligned} \quad (5)$$

and the archimedean kernel $K_\chi^{\mathrm{arch}}(u) := \mathcal{F}^{-1}[\rho_\chi](u)$ with $\rho_\chi(t) = \log(q/\pi) + \Re\, \psi(1/4 + it/2)$, one obtains the kernel $K_\chi(x, y) = \kappa_\chi(x - y) + K_\chi^{\mathrm{arch}}(x - y)$ satisfying $Q_\chi[f] = \iint K_\chi(x, y) f(x) f(y)\, dx\, dy$ on $C_c^\infty([-L, L])$.

C Sector decomposition

Let $\mathcal{P} : H_L \to H_L$, $(\mathcal{P}f)(x) = f(-x)$, denote the reflection. $\mathcal{P}$ is unitary self-adjoint with $\mathcal{P}^2 = I$; the spectral projections $P^\pm = (I \pm \mathcal{P})/2$ split $H_L = H_L^+ \oplus H_L^-$ into even/odd sectors. Standard Fourier bases are

$$e_0^+ = \tfrac{1}{\sqrt{2L}}, \qquad e_n^+(x) = \tfrac{1}{\sqrt{L}} \cos(n\pi x/L),$$
$$e_n^-(x) = \tfrac{1}{\sqrt{L}} \sin(n\pi x/L).$$

The Galerkin truncation at basis size N uses modes $0 \leq n \leq N$ in each sector.

D Sector kernels and the anti-diagonal difference

Because $K_\chi(x, y) = K_\chi(-x, -y)$, the sector-projected kernels take the form

Theorem II.1 (Sector kernels; [6], Thm. 3.2)**.** *On $H_L^\pm$ respectively,*

$$\begin{aligned} K_\chi^\pm(x, y) &= \tfrac{1}{2}\big[\kappa_\chi(x - y) \pm \kappa_\chi(x + y)\big] \\ &\quad + \tfrac{1}{2}\big[K_\chi^{\mathrm{arch}}(x - y) \pm K_\chi^{\mathrm{arch}}(x + y)\big]. \end{aligned}$$

Proof sketch. The reflection $\mathcal{P}f(x) := f(-x)$ commutes with translation T_a via $\mathcal{P}T_a = T_{-a}\mathcal{P}$, and the kernel $K_\chi(x, y) = \kappa_\chi(x - y) + K_\chi^{\mathrm{arch}}(x - y)$ is translation-invariant. Symmetrising under $\mathcal{P}$: $P^\pm K_\chi P^\pm = (K_\chi \pm \mathcal{P}K_\chi\mathcal{P})/2$, where $\mathcal{P}K_\chi\mathcal{P}(x, y) = K_\chi(-x, -y) = K_\chi(x, y)$ by the parity of κ_χ and K_χ^{arch} (both even on $\mathbb{R}$). The off-diagonal symmetrisation $P^\pm K_\chi(1 - P^\pm)$ vanishes on the appropriate sector, leaving the displayed expression. The action on the test-function space $C_c^\infty([-L, L])$ is well-defined because κ_χ is a tempered distribution supported on $\{u = \pm m \log p\}$, all of which lie in $[-2L, 2L]$; pairing with $f \in C_c^\infty([-L, L])$ produces a finite sum over (p, m) with $m \log p \leq 2L$. Full computational details, including the verification that the symmetrisation preserves self-adjointness, are in [6], §3.2. $\square$

Projected eigenvalue problems $K_\chi^\pm \phi_\chi^\pm = \lambda_1(\chi, \pm)\phi_\chi^\pm$ on $H_L^\pm$ define the sector ground states $\phi_\chi^\pm$ which figure centrally below.

Theorem II.2 (Anti-diagonal sector difference; [6], Thm. 4.1)**.** *For an even primitive Dirichlet character χ (i.e., $\chi(-1) = +1$, as throughout this paper),*

$$K_\chi^-(x,y) - K_\chi^+(x,y) = -\kappa_\chi(x+y) - K_\chi^{\mathrm{arch}}(x+y).$$

The primitive part is supported on the anti-diagonals $x + y = \pm m \log p$ and carries the full character dependence; the archimedean part depends on χ only through the constant $\log(q/\pi)$ (in the even-character setting; for odd characters the archimedean kernel acquires an additional $\Re\,\psi(3/4 + it/2)$-type term, which is outside the scope of this atlas).

Proof sketch. By Theorem II.1, $K_\chi^\pm(x,y) = \frac{1}{2}[\kappa_\chi(x-y) \pm \kappa_\chi(x+y)] + \frac{1}{2}[K_\chi^{\mathrm{arch}}(x-y) \pm K_\chi^{\mathrm{arch}}(x+y)]$. Subtracting K_χ^+ from K_χ^- cancels all $(x-y)$-arguments and doubles the $(x+y)$-arguments with sign $-$: $K_\chi^- - K_\chi^+ = -\kappa_\chi(x+y) - K_\chi^{\mathrm{arch}}(x+y)$. For the character-dependence claim: κ_χ depends on χ through every $\chi(p)^m \log p / p^{m/2}$ coefficient ((5)), whereas $K_\chi^{\mathrm{arch}} = \mathcal{F}^{-1}[\rho_\chi]$ with $\rho_\chi(t) = \log(q/\pi) + \Re\,\psi(1/4 + it/2)$ depends on χ only through the constant $\log(q/\pi)$ (the ψ-term is character-independent in the even-only setting). Full derivation: [6], §4.1. ☐

E The gap functional

Let $\lambda_1^{\min}(Q_\chi^\pm)$ denote the smallest eigenvalue of Q_χ restricted to $H_L^\pm$, and write

$$\begin{aligned}\mathrm{gap}_\chi(\lambda) &:= \lambda_1^{\min}(Q_\chi^-) - \lambda_1^{\min}(Q_\chi^+) \\ &= \langle \phi_\chi^- | K_\chi^- | \phi_\chi^- \rangle - \langle \phi_\chi^+ | K_\chi^+ | \phi_\chi^+ \rangle.\end{aligned} \tag{6}$$

For the trivial character (classical Riemann setting), [4] targets a $\sqrt{\lambda}$ sector separation as the even-dominance mechanism; the current cited record frames this as a conditional reduction rather than a closed unconditional estimate. For non-trivial χ, the empirical observation is that $|\,\mathrm{gap}_\chi| = O(1)$ at the accessible λ, with *character-specific* sign; the atlas of Section IV is the quantitative image of this observation.

Remark II.3 (Convention-dependence of sign accuracy)*.* The sign accuracy reported in Table 1 is computed relative to the Connes-operator convention $\tilde{A}_\chi$ (Appendix B) used throughout. Under the equivalent Weil-Q convention Q_χ, all gap signs invert simultaneously (both predictor and reference), so the relative agreement predictor-vs.-reference is unchanged. Thus the sign-accuracy statistic is convention-internal: it tests self-consistency between predictor and reference within a chosen convention, rather than an absolute, convention-free sign of gap_χ. A convention-free sign would require an external arithmetic identification (e.g., from the spectral side via LMFDB), which we do not provide.

III The gap formula: reduced vs. unreduced

A The leading-order reduction

Starting from (6) and Theorem II.2, the gap admits the representation

$$\begin{aligned}\mathrm{gap}_\chi(\lambda) = \langle \phi_\chi^- | K_\chi^- &- K_\chi^+ | \phi_\chi^- \rangle \\ &+ \langle \phi_\chi^- | K_\chi^+ | \phi_\chi^- \rangle - \langle \phi_\chi^+ | K_\chi^+ | \phi_\chi^+ \rangle.\end{aligned} \tag{7}$$

If one approximates both ground states by a common function $\phi_\chi^\pm \approx \phi_\infty$, the second bracket vanishes and the first simplifies, by Theorem II.2, to

$$\begin{aligned}\mathrm{gap}_\chi(\lambda) \overset{?}{\approx} -\iint \phi_\infty(x)\big[\kappa_\chi(x+y) + K_\chi^{\mathrm{arch}}(x+y)\big] \\ \times \phi_\infty(y)\,dx\,dy.\end{aligned} \tag{8}$$

Unfolding κ_χ via (5) and using evenness of ϕ_∞ (even sector) delivers the reduced-gap formula

$$\begin{aligned}\mathrm{gap}_\chi(\lambda) \overset{?}{\approx} 2 \sum_{\substack{p \nmid q \\ m \log p \le 2L}} \frac{\chi(p)^m \log p}{p^{m/2}} \\ \times \Phi_\infty(m \log p) \;+\; \text{arch. term},\end{aligned} \tag{9}$$

with $\Phi_\infty(t) := (\phi_\infty * \phi_\infty)(t)$ (cf. [6], Cor. 4.4 and [8], Cor. 6.1). This is the formula one would naturally transfer from the trivial character case, where the common ground state ϕ_∞ is accessible from the Paley-Wiener analysis of [4].

B Empirical status of the reduced formula

Session 6 of the present programme tested (9) with sixteen candidate windows ϕ_∞: centered Gaussians of width $\sigma \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$, off-center Gaussians, rectangles of cut-off T, $\cos^2$ windows on $[-T, T]$, the raw χ-weighted prime sum, and—most decisively—the *empirical* single-sector ground state $\phi_\infty := \phi_\chi^+$ extracted from the $N = 200$ Galerkin diagonalization.

None of the sixteen candidates reached the success criterion $\ge 9/10$ sign matches with $R^2 \ge 0.8$. The best leading-order Gaussian window achieves 6/10 with $R^2 = 0.15$; the empirical single-sector ground state achieves 4/10 with $R^2 = 0.02$. Details of the 16-window scan are recorded in [9]. The crucial observation is that for all ten characters the empirical single-sector autocorrelation Φ_∞ is nearly identical (numerically within 10^{-3}): the reduction $\phi_\chi^+ \approx \phi_\chi^- \approx \phi_\infty$ is therefore not merely quantitatively inaccurate, but erases the character-specific signal.

Remark III.1 (Where the character signal lives)*.* The sector-restricted ground states are, to leading order, the same function: they are both close to the Paley-Wiener ground state of the diagonal kernel $\kappa_\chi(x-y) + K_\chi^{\mathrm{arch}}(x-y)$. Character-specific information is stored in the difference $\phi_\chi^+ - \phi_\chi^-$, which is the fine structure generated by the anti-diagonal term $\kappa_\chi(x+y)+K_\chi^{\mathrm{arch}}(x+y)$ of Theorem II.2. Any analytic reduction that absorbs ϕ_χ^+ and ϕ_χ^- into a common function loses exactly this signal.

C The unreduced formula

We retain both ground states separately. Define the sector-difference autocorrelation

$$\Delta_\chi(t) := (\phi_\chi^+ * \phi_\chi^+)(t) + (\phi_\chi^- * \phi_\chi^-)(t), \qquad (10)$$

Parity-sign note. The numerical implementation in our scripts (Appendix B) computes the autocorrelation $R_{\phi^\pm}(t) := \int \phi^\pm(s)\phi^\pm(s-t)\,ds$ via `np.correlate`, then forms $\Delta_\chi^{\mathrm{num}}(t) := R_{\phi^+}(t) - R_{\phi^-}(t)$. By the parity identity $R_{\phi^+} = (\phi^+ {*} \phi^+)_{\mathrm{math}}$ (even sector) and $R_{\phi^-} = -(\phi^- {*} \phi^-)_{\mathrm{math}}$ (odd sector), one has

$$\Delta_\chi^{\mathrm{num}}(t) = (\phi^+ {*} \phi^+)_{\mathrm{math}}(t) + (\phi^- {*} \phi^-)_{\mathrm{math}}(t),$$

which is precisely the right-hand side of (10). Thus Δ_χ as defined in (10) agrees identically with the script-computed quantity $\Delta_\chi^{\mathrm{num}}$, and the coefficient -2 in (13) is the unique value that makes the gap identity (Theorem III.2) hold. See Appendix B for the full derivation tracking all six relevant sign sources.

and the archimedean sector difference

$$\mathrm{arch\_diff}_\chi := \langle \phi_\chi^-, K_-^{\mathrm{arch}} \phi_\chi^- \rangle - \langle \phi_\chi^+, K_+^{\mathrm{arch}} \phi_\chi^+ \rangle, \qquad (11)$$

where $K_\pm^{\mathrm{arch}}$ are the sector-restricted archimedean operators (Theorem II.1). Substituting into (7) and using Theorem II.2 yields the unreduced formula stated below.

Theorem III.2 (Unreduced gap formula)**.** *Let χ be a primitive real character of conductor q with $\chi(-1) = +1$, and let $\lambda > 1$, $L = \log \lambda$. Let $\phi_\chi^\pm \in H_L^\pm$ be the sector ground states of $K_\chi^\pm$, each L^2-normalized. Then*

$$\mathrm{gap}_\chi(\lambda) = S_\chi(\lambda) + \mathrm{arch\_diff}_\chi, \qquad (12)$$

with

$$S_\chi(\lambda) = -2 \sum_{\substack{p \nmid q \\ m \log p \le 2L}} \frac{\chi(p)^m \log p}{p^{m/2}} \Delta_\chi(m \log p). \qquad (13)$$

A derivation is given in [6], Remark 4.5 (with sign convention reconciled against `build_W`; see Appendix B). At the level of the *Galerkin truncation* at basis size N, the equation (12) also holds exactly:

$$\mathrm{gap}_{\mathrm{gal}}^{(N)}(\chi) = S_\chi^{(N)}(\lambda) + \mathrm{arch\_diff}_\chi^{(N)}, \qquad (14)$$

provided $\phi_\chi^\pm$ are taken as the ground-state eigenvectors of the Galerkin-truncated operators $K_\chi^{\pm,N}$ at basis size N and $S^{(N)}$, $\mathrm{arch\_diff}^{(N)}$ are computed from these finite-N data.

D Two numerical realizations of $S_\chi^{(N)}$

A subtle point that drives all of what follows is that (13) admits *two* distinct numerical implementations from Galerkin data, which agree to within 10^{-3} but not identically.

(R1) *Spatial autocorrelation.* Evaluate $\phi_\chi^\pm$ on a grid, compute $(\phi^\pm * \phi^\pm)(t)$ with a numerical correlation, interpolate to the prime abscissae $t = m \log p$, and sum.

(R2) *Matrix decomposition.* Decompose the Galerkin matrix $W^\pm = W_{\mathrm{arch}}^\pm + W_{\mathrm{prime}}^\pm$ and form $\mathrm{prime\_diff}_{\mathrm{matrix}} := \langle \phi^-, W_{\mathrm{prime}}^- \phi^- \rangle - \langle \phi^+, W_{\mathrm{prime}}^+ \phi^+ \rangle$.

Both quantities estimate $S_\chi^{(N)}$; (R1) is obtained via interpolated autocorrelation, (R2) via exact matrix inner product. Their discrepancy is at the level of floating-point and interpolation error, of order 10^{-3}. For characters with $|\,\mathrm{gap}_\chi\,| \gtrsim 10^{-2}$ this is negligible. For the weakly signalled characters χ_{21}, χ_{29}, χ_{60}, with $|\,\mathrm{gap}\,| \lesssim 10^{-2}$, it is of the same order as the signal. Section VI is devoted to the consequences.

IV Numerical validation

A Computational setup

For each of the ten real characters χ_D, $D \in \{5, 8, 12, 13, 17, 21, 24, 29, 33, 60\}$, we construct the Galerkin matrices $W_\chi^\pm$ in the Fourier bases of $H_L^\pm$ at cutoff $\lambda = 20000$ ($L = \log \lambda \approx 9.90$). Matrix assembly uses the explicit-formula coefficients $\chi(p^m) \log p / p^{m/2}$ for all prime powers with $m \log p \le 2L$ and the archimedean operator K_χ^{arch} computed by Fast Fourier transform of ρ_χ.

Three Galerkin resolutions are used: $N = 200$ locally on a desktop, $N = 400$ locally, and $N = 600$ on an 8 GB Hetzner CCX13 cloud instance (total wall-clock 1.95 hours across all ten characters). The smallest eigenvalue of each $W_{\chi,N}^\pm$ is obtained by a Lanczos inverse iteration with tolerance 10^{-10}, producing the sector ground state $\phi_\chi^{\pm,(N)}$ and the Galerkin gap $\mathrm{gap}_{\mathrm{gal}}^{(N)}(\chi) := \lambda_1^{\min}(W_{\chi,N}^-) - \lambda_1^{\min}(W_{\chi,N}^+)$.

For every character, the quantities $S_\chi^{(N)}$ (via spatial-autocorrelation realization (R1)) and $\mathrm{arch\_diff}_\chi^{(N)}$ are evaluated from the ground states, and the sum $S_\chi^{(N)} + \mathrm{arch\_diff}_\chi^{(N)}$ is compared to $\mathrm{gap}_{\mathrm{gal}}^{(N)}(\chi)$ and to the high-N Galerkin reference gap $\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}(\chi)$ with $N_{\mathrm{src}} \in \{400, 600\}$ (cf. §B). Scripts `arch_term_analysis.py` and `server_n600_full_analysis.py` record the raw outputs in `ARCH_TERM_ANALYSIS.json` and `ARCH_TERM_N600_SERVER.json` respectively; post-processing tables are in `ARCH_TERM_ANALYSIS.md` and `ARCH_TERM_N600_ANALYSIS.md`.

Table 1: Sign and linear-regression performance of the predictor realizations against the high-N Galerkin reference gap $\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}(\chi_D)$ at $\lambda = 20000$, $N \in \{200, 600\}$. Sign accuracy is computed relative to the Connes-convention sign of the high-N Galerkin reference gap (sign-convention note below the table). * N=200 statistics are subject to the Galerkin-truncation effects documented in Remark IV.1 and §C; the N=600 row is the operationally preferred result. Excluding χ_{21} (the single sign-reversal outlier; see §VII) yields 9/9 sign accuracy for $\mathrm{gap}_{\mathrm{gal}}$ at $N = 600$ (and 8/9 for $S + \mathrm{arch\_diff}$), but exclusion is by construction—the excluded character is precisely the failure case—and therefore not reported as an independent statistic.

Configuration	N	sign_ok	R^2
$\mathrm{gap}_{\mathrm{gal}}$ (all 10)*	200	9/10	0.41
$S + \mathrm{arch\_diff}$ (all 10)*	200	9/10	0.42
$\mathrm{gap}_{\mathrm{gal}}$ (all 10)	600	10/10	0.99
$S + \mathrm{arch\_diff}$ (all 10)	600	8/10	0.95

Sign-convention note. Sign accuracy is relative to the Connes-operator convention $\tilde{A}_\chi$ used throughout (Appendix B). The opposite convention (Weil-Q, coefficient $+2$) would invert all signs and yield a complementary sign-accuracy statistic, but the relative agreement predictor-vs.-reference is unchanged.

B The central result table

Table 2 summarizes the comparison at $\lambda = 20000$ for all ten characters. For each character we list the high-N Galerkin reference gap $\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}$ (with the resolution N_{src} from which it is drawn), the Galerkin gap at $N = 200$ and $N = 600$, and the predictor $S + \mathrm{arch\_diff}$ at both resolutions. Two observations anchor the entire analysis.

First, at $N = 600$ the entries labelled "$\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}$" and "$\mathrm{gap}_{\mathrm{gal}}^{(600)}$" coincide identically whenever the reference was itself computed at $N = 600$ (the six characters $\chi_8, \chi_{13}, \chi_{21}, \chi_{29}, \chi_{33}, \chi_{60}$); this is *not* a verification of the formula, but a tautology exposed in §VI.

Second, the predictor $S^{(600)} + \mathrm{arch\_diff}^{(600)}$ achieves high linear correlation $R^2 = 0.95$ against the empirical gaps but drops to 8/10 sign accuracy because of two sign flips at χ_{21} and χ_{29}.

C Regression statistics

We record all four canonical regression configurations in Table 1. The pattern is pivotal for what follows.

At $N = 200$, the predictor $S + \mathrm{arch\_diff}$ reaches 9/10 sign accuracy ($R^2 = 0.42$ on the full sample, $R^2 = 0.80$ on the subsample excluding χ_{21} post hoc—a by-construction statistic, not reported as an independent test; cf. caption of Table 1). This apparent numerical closure motivated the "Weil-kernel closure paper" title during early Session 7 of the development. At $N = 600$, however, three things happen simultaneously: (i) the Galerkin gap $\mathrm{gap}_{\mathrm{gal}}^{(600)}$ becomes numerically indistinguishable from the empirical reference for six of ten characters (tautology, §VI); (ii) χ_{21}'s Galerkin gap collapses from $+0.282$ to -0.004, confirming it as an N-oscillator (§VII); and (iii) the predictor $S + \mathrm{arch\_diff}^{(600)}$ gains one point on R^2 but loses two points on sign accuracy, now failing both at χ_{21} (where the $N = 600$ Galerkin gap is small) and at χ_{29} (where numerical realization (R1) diverges in sign from realization (R2)).

Remark IV.1 (Siegel-Walfisz compression at $N = 600$). Comparing the $\mathrm{gap}_{\mathrm{gal}}$-columns at $N = 200$ and $N = 600$ reveals a systematic compression effect: for the strong-positive character χ_{12}, $\mathrm{gap}_{\mathrm{gal}}$ falls from $+0.0325$ to $+0.0115$ (factor 0.35); for χ_{17}, χ_{24}, χ_{29}, χ_{60} the same shrinkage factor holds to within ± 0.1. The strong-negative characters χ_8, χ_{13}, χ_{33} are qualitatively stable between $N = 200$ and $N = 600$ (relative change $\lesssim 10\%$). This dichotomy is qualitatively reminiscent of the Siegel-Walfisz upper bound $|\sum_{p \le x} \chi(p) \log p| = o(\sqrt{x})$ applied to the large-prime tail—characters with a genuine low-conductor signal appear largely determined already at $N = 200$, whereas weakly-signalled characters have their $N = 200$ estimates compressed by a factor approximately 0.35 at $N = 600$—but no theoretical derivation linking the compression factor to a Siegel-Walfisz constant is offered here. We flag the empirical observation as an open question; see §IX, point on the theoretical derivation of the Siegel-Walfisz compression.

D Evidence-layer reading

The numerical presentation should be read as a layered diagnostic rather than as a single validation statistic. Table 3 spells out which inference each layer can support and which inference remains outside the present evidence base.

V Character atlas in four axes

The raw table of Section IV admits a structural reading along four taxonomic dimensions: two *filter constants* (parity and root number, both equal to $+1$ across the entire sample) and two *informative axes* (archimedean weight ratio and gap signature). We develop each dimension in turn, in order of increasing informativeness.

A Parity cartography

The parity $\chi(-1) \in \{+1, -1\}$ decides the archimedean factor of $\Lambda(s, \chi)$: even characters use $\Gamma(s/2)$, odd characters use $\Gamma((s+1)/2)$. In this paper all ten characters are Kronecker characters χ_D with positive discriminant D, and hence all even. This is deliberate: the sector decomposition $H_L = H_L^+ \oplus H_L^-$ used throughout assumes even parity, which ensures a non-trivial cosine-sine asymmetry of the anti-diagonal kernel of Theorem II.2 (odd characters would require the paired-space formalism $\chi \oplus \bar{\chi}$ or a tensor with a sign twist; cf. [5], §5.5, footnote on Gamma parity). The even restriction thus fixes the qualitative regime and leaves room for the three finer classifications below.

Table 2: N-convergence of the Galerkin gap and the Weil-kernel predictor at $\lambda = 20000$ for ten low-conductor real characters. Note that $\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}$ entries with $N_{\mathrm{src}} = 600$ coincide identically with $\mathrm{gap}_{\mathrm{gal}}^{(N=600)}$ because the reference values are themselves outputs of the same Galerkin diagonalization at $N = 600$; this is the tautology discussed in §VI. There is no external, convention-free reference here—all gap values originate from the Galerkin method itself; cf. §B. Sources: `ARCH_TERM_ANALYSIS.md`, `ARCH_TERM_N600_ANALYSIS.md`.

χ	D	$\mathrm{gap}_{\mathrm{gal}}^{(N_{\mathrm{src}})}$	$\mathrm{gap}_{\mathrm{gal}}^{(200)}$	$\mathrm{gap}_{\mathrm{gal}}^{(600)}$	$(S + \mathrm{arch\_diff})^{(200)}$	$(S + \mathrm{arch\_diff})^{(600)}$
χ_5	5	+0.01560 (400)	+0.01429	+0.01603	$+3.35 \cdot 10^{-2}$	$+3.37 \cdot 10^{-2}$
χ_8	8	−0.04902 (600)	−0.04548	−0.04902	$-4.22 \cdot 10^{-2}$	$-4.08 \cdot 10^{-2}$
χ_{12}	12	+0.03367 (400)	+0.03251	+0.01154	$+5.07 \cdot 10^{-2}$	$+2.03 \cdot 10^{-2}$
χ_{13}	13	−0.12504 (600)	−0.12502	−0.12504	$-1.15 \cdot 10^{-1}$	$-1.16 \cdot 10^{-1}$
χ_{17}	17	+0.01318 (400)	+0.00408	+0.00886	$+1.67 \cdot 10^{-2}$	$+1.13 \cdot 10^{-2}$
χ_{21}	21	−0.00423 (600)	**+0.28163**	−0.00423	$+2.89 \cdot 10^{-1}$	$+1.14 \cdot 10^{-2}$ **(F)**
χ_{24}	24	+0.01076 (400)	+0.01632	+0.00866	$+1.74 \cdot 10^{-2}$	$+1.86 \cdot 10^{-2}$
χ_{29}	29	+0.01439 (600)	+0.02372	+0.01439	$+2.58 \cdot 10^{-2}$	$-1.16 \cdot 10^{-2}$ **(F)**
χ_{33}	33	−0.14221 (600)	−0.14051	−0.14221	$-1.31 \cdot 10^{-1}$	$-1.32 \cdot 10^{-1}$
χ_{60}	60	+0.00521 (600)	+0.12363	+0.00521	$+1.27 \cdot 10^{-1}$	$+7.21 \cdot 10^{-3}$
$S + \mathrm{arch\_diff}$ *sign accuracy ($N = 200$):*					9/10 (R^2=0.42)	
$S + \mathrm{arch\_diff}$ *sign accuracy ($N = 600$):*						8/10 (R^2=0.95)
$\mathrm{gap}_{\mathrm{gal}}$ *sign accuracy ($N = 600$):*						10/10 (R^2=0.99, tautology)

Table 3: Evidence-layer ledger for the numerical atlas. The table records what each data layer can legitimately support and which inference remains outside the present evidence base.

Layer	Data anchor	Supports	Does not support
Raw gap layer	Table 2 and the source reports listed there	High-N Galerkin self-consistency and the localization of sign-sensitive characters such as χ_{21}, χ_{29} and χ_{60}.	External truth of the $N \to \infty$ gap or closure of $C_\chi^{(2)}$.
Regression layer	Table 1	The $N = 200$ apparent closure is truncation-sensitive, while the $N = 600$ predictor has high R^2 but two sign failures.	A universal sign rule or a convention-free validation statistic.
Atlas-taxonomy layer	Table 4 and §D	Sample-specific prime/mixed/arch regimes and gap-signature classes.	Universality beyond the ten low-conductor characters.
Failure-control layer	Proposition VI.1 and Table 5	The matrix-decomposition identity and the χ_{21} oscillator block a closure interpretation of the Galerkin numerics.	A replacement analytic ground-state theory.
Missing-reference layer	§B	Identifies the external gates needed before predictive claims: higher precision Galerkin, LMFDB-zero comparison or an independent basis.	Any completed external validation in the present atlas.

B Root-number classification

For every real Dirichlet character the root number $W(\chi) := \epsilon_\chi$ is +1 by the classical Gauss-sum formula $\tau(\chi) = \sqrt{q}$ for primitive real even characters [11]. All ten characters in our atlas thus share the same root number $W = +1$, and root number is constant across the table. Consequently root number cannot discriminate gap signs within our sample. A larger sample extending to complex characters (§IX) would exhibit a genuine root-number axis with $W \in \{+1, -1, \pm i\}$; we flag this as part of the open outlook rather than as a result of the present atlas.

C Archimedean-factor layer

The quantity $r_\chi := |\mathrm{arch\_diff}_\chi^{(200)}|/|S_\chi^{(200)}|$ (ratio of archimedean to prime contribution at $N = 200$, $\lambda = 20000$) varies by nearly two orders of magnitude across the sample. Table 4 records the values.

Three sub-classes emerge:

- *Prime-dominated* ($r_\chi \leq 25\%$): χ_{12}, χ_{13}, χ_{17}, χ_{21}, χ_{24}, χ_{29}, χ_{33}. The prime sum S_χ determines both magnitude and sign of the gap; $\mathrm{arch\_diff}_\chi$ shifts the quantitative value by no more than a quarter. This is the "natural" regime in which intuition about the Weil kernel carries.
- *Mixed* ($r_\chi \in [60\%, 70\%]$): χ_5, χ_8. Primes and archimedean factor are comparable; neither alone is a reliable predictor.

Table 4: Archimedean weight ratio $r_\chi = |\,\text{arch\_diff}\,|/|S|$ at $N = 200$, $\lambda = 20000$.

χ_D	$\lvert S_\chi \rvert$	$\lvert\,\text{arch\_diff}_\chi\,\rvert$	r_χ	Class
χ_{29}	$2.54\cdot10^{-2}$	$4.47\cdot10^{-4}$	1.8%	prime-dominated
χ_{24}	$1.62\cdot10^{-2}$	$1.19\cdot10^{-3}$	7.4%	prime-dominated
χ_{33}	$1.65\cdot10^{-1}$	$3.44\cdot10^{-2}$	21%	prime-dominated
χ_5	$9.06\cdot10^{-2}$	$5.71\cdot10^{-2}$	63%	mixed
χ_8	$1.16\cdot10^{-1}$	$7.43\cdot10^{-2}$	64%	mixed
χ_{13}	$1.02\cdot10^{-1}$	$1.33\cdot10^{-2}$	13%	prime-dominated
χ_{17}	$2.02\cdot10^{-2}$	$3.52\cdot10^{-3}$	17%	prime-dominated
χ_{12}	$6.51\cdot10^{-2}$	$1.44\cdot10^{-2}$	22%	prime-dominated
χ_{21}	$2.31\cdot10^{-1}$	$5.77\cdot10^{-2}$	25%	prime-dominated†
χ_{60}	$4.95\cdot10^{-2}$	$1.76\cdot10^{-1}$	356%	arch-dominated

† at $N = 200$; the effective weights shift at $N = 600$ as the Galerkin ground states settle (cf. §VII).

- *Arch-dominated* ($r_\chi > 100\%$): χ_{60}. Here the archimedean sector difference *exceeds* the prime contribution by a factor ~ 4, and S_χ and arch_diff_χ even disagree in sign ($S = -4.95\cdot10^{-2}$, $\text{arch\_diff} = +1.76\cdot10^{-1}$). The reference gap $\text{gap}_{\text{gal}}^{(600)} = +0.005$ is near zero and is carried entirely by the archimedean term.

The sub-class distinction is structural, not just quantitative: for prime-dominated characters a future analytic control of $\Delta_\chi(m \log p)$ would translate directly into gap-sign prediction, whereas for arch-dominated characters the archimedean ψ-function asymptotics dominate and an independent analysis is needed.

Remark V.1 (Descriptive vs. predictive status of the trichotomy). With ten characters, the prime/mixed/arch trichotomy of Table 4 is descriptive of the sample, not predictive for arbitrary primitive real characters: the threshold values ($r \le 25\%$, $r \in [60\%, 70\%]$, $r > 100\%$) are chosen to separate the observed clusters, with no datapoint in the gap intervals $r \in (25\%, 60\%)$ and $r \in (70\%, 100\%)$. Whether the trichotomy generalises is a hypothesis to be tested on the conductor extension $D \le 100$ (§IX, point 4); the present atlas does not assert it as a universal classification.

D Gap-signature atlas

The four classes below organize the ten characters by the quantitative nature of their gaps, with the high-N Galerkin reference $\text{gap}_{\text{gal}}^{(N_{\text{src}})}$ read off Table 2.

Strong-negative prototype. χ_{33} is the outstanding case: $\text{gap}_{\text{gal}}^{(600)} = -0.142$, with near-identical values at $N \in \{200, 400, 600\}$. Both prime contribution and empirical gap are consistently negative, and the gap is the largest in absolute value over the sample. The character is $\chi_{33}(n) = \left(\frac{33}{n}\right)$, with factorization $33 = 3\cdot 11$; the corresponding Legendre-symbol values at small primes are $\chi_{33}(2, 5, 7, 13, 17, 19, 23, 29, 31, 37) = (1, -1, -1, -1, -1, -1, +1, -1, +1, -1)$, delivering a clear bias towards -1. We designate χ_{33} the *strong-negative prototype* of the atlas.

Strong-positive prototype. χ_{12} is the counterpart on the positive side: $\text{gap}_{\text{gal}}^{(400)} = +0.034$, compressed to $\text{gap}_{\text{gal}}^{(600)} = +0.012$ at $N = 600$ (Remark IV.1). The primitive character mod 12 has conductor a product of two small primes ($q = 12 = 4 \cdot 3$); earlier asymptotic studies [10] observed a slope consistent with zero across three orders of magnitude in λ at $N = 200$, which we now reinterpret as an N-limited observation (see also `THEORIE_CHI12_KONSTANZ.md`). Within the present atlas χ_{12} is the strong-positive prototype, albeit at smaller magnitude than χ_{33}.

Moderate-negative class. χ_8 and χ_{13} both exhibit stable negative gaps (-0.049 and -0.125 respectively) with consistent signs across $N \in \{200, 400, 600\}$. They lie between the strong-negative prototype and the weak-positive cluster.

Weak-positive cluster. χ_{17}, χ_{24}, χ_{29} form a tight cluster of small positive gaps ($+0.013$, $+0.011$, $+0.014$). Their absolute values lie within the Siegel-Walfisz compression regime; their predictor $S + \text{arch\_diff}$ retains sign at $N = 200$ but can flip at $N = 600$ (case of χ_{29}, realization (R1); cf. §C). This cluster marks the lower bound of reliable numerical sign resolution for our current methods.

Arch-dominated near-zero case. χ_{60} has $\text{gap}_{\text{gal}}^{(600)} = +0.005$ with $r_{\chi_{60}} = 356\%$ (above). The predictor $S + \text{arch\_diff}$ happens to have the correct sign at $N = 600$, but not as a consequence of the prime contribution, which disagrees in sign with the empirical gap; instead the archimedean term dominates. We flag χ_{60} as a distinct regime warranting a separate analysis.

Composite-conductor oscillator. χ_{21} ($D = 21 = 3 \cdot 7$) has $\text{gap}_{\text{gal}}^{(600)} = -0.004$ but an $N = 200$ Galerkin gap of $+0.282$ — a sign flip of factor 66. It is the only character in our sample exhibiting this behavior, and we treat it in full detail in §VII.

The four-axis cartography above is the paper's descriptive payoff: it groups the ten characters into qualitatively distinct regimes before, and independently of, any predictive formula. In particular the prime-dominated/mixed/arch-dominated trichotomy (Table 4) and the gap-signature classes (strong-neg, strong-pos, moderate-neg, weak-pos, arch-dominated, oscillator) together exhibit six sub-classes in ten characters—a first sign that the "single-axis transfer" of the classical even-dominance regime [4] is too coarse.

VI The matrix-decomposition tautology

Proposition VI.1 (Tautology). *Let $\phi_\chi^\pm$ be the ground-state eigenvectors of the Galerkin-truncated operators*

$K_\chi^\pm$ at basis size N. Then for any decomposition $W^\pm = W^\pm_{\mathrm{arch}} + W^\pm_{\mathrm{prime}}$ of the Galerkin matrix,

$$\mathrm{gap}^{(N)}_{\mathrm{gal}} = \mathrm{arch\_diff} + \mathrm{prime\_diff}_{\mathrm{matrix}},$$
$$\mathrm{arch\_diff} := \langle \phi^-, W^-_{\mathrm{arch}}\phi^- \rangle - \langle \phi^+, W^+_{\mathrm{arch}}\phi^+ \rangle,$$
$$\mathrm{prime\_diff}_{\mathrm{matrix}} := \langle \phi^-, W^-_{\mathrm{prime}}\phi^- \rangle - \langle \phi^+, W^+_{\mathrm{prime}}\phi^+ \rangle.$$

Proof. Linearity of the inner product on the Galerkin space gives $\langle \phi^\pm, W^\pm\phi^\pm\rangle = \langle \phi^\pm, W^\pm_{\mathrm{arch}}\phi^\pm\rangle + \langle \phi^\pm, W^\pm_{\mathrm{prime}}\phi^\pm\rangle$. Since $\phi^\pm$ are eigenvectors with eigenvalue $\lambda_1^{\min}(W^\pm)$, the gap $\mathrm{gap}^{(N)}_{\mathrm{gal}}$ decomposes as the difference of the two right-hand sides. □ □

Although the propositional content is a trivial linearity identity, its *methodological consequence* is non-trivial: any apparent agreement between $\mathrm{gap}^{(N)}_{\mathrm{gal}}$ and $S_\chi + \mathrm{arch\_diff}_\chi$ at fixed N is structural rather than predictive, not a prediction. When $\mathrm{prime\_diff}_{\mathrm{matrix}}$ is replaced by the spatial-domain autocorrelation formula S_χ, a small numerical discrepancy arises (of order 10^{-3}, about 1–2% of S); the "9/9 sign accuracy at $N = 200$" reported in preliminary work of the present author admits two equivalently consistent interpretations: (a) the discrepancy was *fortuitously small* at low N for the sample of ten characters and grows at higher N as the predictor's intrinsic limitations become resolved; (b) at $N = 600$ a *conditioning pathology* sets in for characters with $|\,\mathrm{gap}\,| \lesssim 10^{-2}$, where floating-point and interpolation errors of order 10^{-3} become sign-relevant. The two readings cannot be distinguished without an external reference (cf. §B). In either case, the $N = 200$ agreement is not a structural property of the formula. At $N = 600$ the discrepancy becomes sign-relevant for χ_{29} (see Table 2).

Remark VI.2 (Why autocorrelation-numerical S fails). The autocorrelation $(\phi^+_\chi * \phi^+_\chi)(t)$ requires numerical evaluation on a discrete grid, combined with interpolation to prime abscissae $t = m \log p$. Floating-point and interpolation errors at the grid scale $\Delta x = 2L/n_{\mathrm{grid}}$ translate into errors of order 10^{-3} in S, which is the same order of magnitude as $|\,\mathrm{gap}\,|$ for weakly-signaled characters like χ_{21}, χ_{29}.

Corollary VI.3 (Where predictive power must come from)**.** *A genuinely predictive Weil-kernel gap formula requires analytic control of the ground states $\phi^\pm_\chi$ without passing through Galerkin diagonalization. The route is Paley-Wiener asymptotic analysis (à la Connes [14], adapted to the Dirichlet setting) or a direct spectral identification of $\Delta_\chi(t)$ from the local data of χ (parity, root number, conductor, arithmetic factor of the L-function).*

VII The χ_{21} N-oscillation

The character χ_{21}, with composite conductor $q = 21 = 3 \cdot 7$, is the only entry of Table 2 where the Galerkin gap at $N = 200$ and the empirical reference at $N = 600$ disagree in sign. We quantify the discrepancy and attribute it to a Galerkin-truncation effect that is resolved at high N, not to a defect of the formula.

Table 5 records the N-convergence data at $\lambda = 20000$, from the script `chi21_n_convergence.py`.

Table 5: N-convergence of χ_{21} at $\lambda = 20000$. The Galerkin gap is flat between $N = 200$ and $N = 400$ and then jumps by more than an order of magnitude between $N = 400$ and $N = 600$. For comparison the strong-positive character χ_{12} converges monotonically (last column).

N	$\mathrm{gap}_{\mathrm{gal}}(\chi_{21})$	$S + \mathrm{arch\_diff}$	$\mathrm{gap}_{\mathrm{gal}}(\chi_{12})$
200	+0.2816	+0.2888	+0.0325
400	+0.2791	+0.2869	+0.0337
600	−0.0042	+0.0114	+0.0115

Two observations structure the analysis.

(i) *Step discontinuity.* Between $N = 200$ and $N = 400$ the Galerkin gap is numerically stationary ($0.2816 \to 0.2791$, change 0.9%); it then drops by a factor of ~ 66 in absolute value and flips sign between $N = 400$ and $N = 600$. This is not the signature of a smooth $N^{-\alpha}$ convergence; the data are consistent with—but do not unambiguously prove—a *suspected quasi-degeneracy collapse* (full convergence verified only up to $N = 600$) in which the Galerkin-truncated sector eigenvalues $\lambda_1^{\min}(W^\pm_{\chi_{21},N})$ for $N \in \{200, 400\}$ correspond to a pseudo-ground state distinct from the infinite-N ground state, with the true ground state becoming representable only once the basis is large enough. A definitive diagnosis would require spectral data $\lambda_2 - \lambda_1$ and eigenvector overlap $|\langle \phi^{(N=200)}, \phi^{(N=600)}\rangle|$ at intermediate $N \in \{500, 700, 800\}$; we flag this as the most direct further verification (cf. §IX, point on χ_{21} spectral diagnosis).

(ii) *Predictor tracks the collapse.* The predictor $S + \mathrm{arch\_diff}$ is dominated by the ground-state autocorrelation and therefore *also* collapses between $N = 400$ and $N = 600$, from $+0.287$ to $+0.011$. At $N = 600$ the predictor no longer agrees in sign with the empirical gap, but its magnitude is now consistent with the empirical magnitude to within a factor ~ 3. This confirms that the formula itself is not anomalous: once the Galerkin ground state has converged, the predictor converges alongside it, with sign determined by the near-zero numerical residuals.

For contrast, χ_{12} (Table 5, last column) shows a monotone N-convergence with no sign flip: $+0.0325 \to +0.0337 \to +0.0115$. This is the generic behaviour on the stable characters. The discontinuity for χ_{21} is therefore neither a Galerkin artefact endemic to composite-conductor characters in general (since χ_{12}, with $q = 12 = 4 \cdot 3$, is composite), nor a systematic boundary of the method, but a suspected character-specific near-degeneracy whose definitive spectral analysis (via $\lambda_2 - \lambda_1$ scan and eigenvector overlaps) lies beyond the scope of the present atlas; see §IX.

Remark VII.1 (Not a formula defect). The sign flip in Table 5 is a property of the Galerkin operator $W^{\pm}_{\chi_{21},N}$ at low N, not of the formula (12). Since $S + \text{arch\_diff}$ is defined in terms of the Galerkin eigenvectors, any spurious eigenvalue at low N is inherited by the predictor. The right theoretical object is $\text{gap}^{(600)}_{\text{gal}}(\chi_{21}) = -0.00423$, and the right question about χ_{21} is whether a Paley-Wiener analysis of the infinite-N ground state would recover this value directly, without passing through the $N = 200, 400$ plateau.

VIII Discussion

A Implications for the three-component decomposition

The three-component decomposition (1) divides any RH-type question into the Galerkin bridge GBC, the Connes-style operator-identification component $C^{(2)}$, and the Hurwitz reality-of-zeros transfer. For the trivial character, [4] isolates the even-dominance route to $C^{(2)}_{\chi=\zeta}$ via the target $\sqrt{\lambda}$ separation of the sectors; in its current Zenodo version this is a conditional reduction with explicit remaining inputs. For non-trivial real χ of small conductor, our atlas demonstrates three structural facts about $C^{(2)}_{\chi}$:

(I1) The Weil-kernel architecture (Theorems II.1 and II.2) is rigorous and transferable. The anti-diagonal sector difference carries the full character dependence, as predicted.

(I2) At $\lambda = 20000$ (a single λ-value), the gap $\text{gap}_{\chi}(\lambda)$ is bounded in absolute value by 0.15 across all ten characters, with character-specific sign. A full λ-asymptotic ($\Omega(\sqrt{\lambda})$ vs. $O(1)$ vs. intermediate growth) cannot be inferred from a single λ-measurement and is left for future work; preliminary results for χ_5 and χ_{12} over three orders of magnitude in λ at $N = 200$ are consistent with $O(1)$ growth ([10], supplementary), but a systematic λ-scan over all ten characters is open. The coarse "universal transfer" hypothesis "even dominance $\Rightarrow$ even sector wins for all χ" is refuted at $\lambda = 20000$ (since six of ten characters have negative or near-zero gap); the λ-asymptotic refutation requires the dedicated λ-scan above.

(I3) Identifying $C^{(2)}_{\chi}$ requires analytic control of the sector ground states $\phi^{\pm}_{\chi}$ themselves, not merely of their eigenvalues. Galerkin diagonalization delivers the eigenvalues but, by the tautology of Proposition VI.1, produces no predictive information that does not enter the computation as input.

In the meta-programme language of [5]: for non-trivial χ, $C^{(2)}_{\chi}$ is a genuinely open component, and its closure is the central technical obstacle to a Hilbert-Pólya-style proof of GRH in the Dirichlet case. The analytic architecture here is ready to accept such a closure; what is missing is the χ-specific asymptotic analysis of $\phi^{\pm}_{\chi}$.

After the Zookeeper paper [3], this missing component can be phrased more sharply in CCM language. The Riemann cell now has an independent Zookeeper kernel package—rank-one decomposition, Cauchy interlacing, and three-lemma closure—beside the v2.1 even-dominance package. The natural Dirichlet back-transfer is therefore Route D: construct a χ-twisted CCM operator with a conductor rank-one term and a χ-weighted arithmetic perturbation, then compare its defect norm to the atlas control quantities. In this reading the Paley-Wiener failures above are not a dead end; they identify the first concrete stress test for the χ-twisted CCM route.

B Limitation: absence of an external reference

All gap values reported in Tables 2, 1, and 4 originate from the same Galerkin diagonalization, evaluated at $N \in \{200, 400, 600\}$. There is therefore *no external, convention-free reference* in this atlas: the high-N values labelled $\text{gap}^{(N_{\text{src}})}_{\text{gal}}$ are the highest available Galerkin estimates, not measurements of the true infinite-N gap. Statistical agreement between the predictor $S_{\chi}+\text{arch\_diff}_{\chi}$ and $\text{gap}^{(N_{\text{src}})}_{\text{gal}}$ is therefore a self-consistency check within the Galerkin method, not an external validation. A genuine external reference would require one of:

(i) mp-arithmetic Galerkin computation at $N \geq 2000$ to diagnose finite-precision artefacts;

(ii) direct spectral evaluation via the LMFDB zeros of $L(s, \chi_D)$;

(iii) an independent Galerkin scheme using a different basis (e.g., Hermite functions instead of Fourier).

None of these is provided here; we flag this as a key open methodological item before any predictive claim about $\text{gap}_{\chi}(\lambda)$ at the present λ can be made.

C The slope-0.72 regression and its meaning

A secondary quantitative fact exposed by our data is the slope of the linear regression of $\text{gap}^{(N_{\text{src}})}_{\text{gal}}$ against $S + \text{arch\_diff}$ at $N = 200$: $\text{gap}^{(N_{\text{src}})}_{\text{gal}} \approx 0.72 \cdot (S + \text{arch\_diff}) - 0.01$ (stable-9 subsample, $R^2 = 0.80$, observational only; cf. §B). We recorded this in Session 7 of the development with enthusiasm, but subsequent reflection shows its content to be modest. The slope measures the multiplicative mismatch between the $N = 200$ predictor $S^{(200)} + \text{arch\_diff}^{(200)}$ and the empirical high-N gap. Under the Siegel-Walfisz compression of Remark IV.1, this mismatch disappears at higher N: at $N = 600$ the Galerkin gap *is* the empirical reference (by identity, not by prediction), so the slope tends to 1 by construction. The 0.72 is therefore an $N = 200$-specific Galerkin-truncation constant, not a theoretical invariant of the formula.

A genuine theoretical analogue of the slope would be the subleading N-dependence of the Galerkin ground states $\phi^{\pm}_{\chi,N}$, controlled by the Paley-Wiener roadmap of [4] adapted to Dirichlet L-functions (cf. [8], §3, and the REG-λ-conditioned rate of [4], §6). We flag this as an open quantitative question of independent interest.

Remark VIII.1 (Status of N=200 statistics). The interpretation of the slope-0.72 as an $N = 200$-specific truncation constant has methodological consequences for the rest of the atlas: all statistics reported at $N = 200$ in Tables 2, 1, and 4 are subject to the same finite-truncation effect. Where the $N = 600$ value differs from the $N = 200$ value, the $N = 600$ value is operationally preferred. The $N = 200$ data is retained for transparency of the convergence behaviour, not as a final result. In particular, the archimedean weight ratios r_χ in Table 4 are computed at $N = 200$ because the matrix-decomposition $W = W_{\text{arch}} + W_{\text{prime}}$ is most cleanly resolved at this resolution; their values would shift modestly at $N = 600$, but the prime/mixed/arch trichotomy is qualitatively preserved (verification: cf. supplementary material, `ARCH_TERM_N600_ANALYSIS.md`).

D Sign convention: rigorous algebraic derivation

The coefficient -2 in (13) is determined algebraically by the structure of the Weil quadratic form (Eq. (4)): the prime-side contribution carries the factor $-2\sum_{p,m} w_{p,m} R_\phi(m \log p)$, and the sector gap inherits this factor via the linearity of the inner product on $H_L^{\pm}$ and the sector eigenvector expansion. The full derivation, tracking all six relevant sign sources (Weil-formula prime-side minus, operator convention, gap orientation, parity action on basis, convolution-vs-autocorrelation parity identity, and the real-character absorption factor), is given in Appendix B and [7], Theorem 5.1.

The discovery sequence in our development scripts—an initial implementation with coefficient $+2$ producing $R = -0.70$ on the sample, corrected to -2 producing $R = +0.70$ on the same sample—reflects an early implementation pitfall in the autocorrelation parity convention (the sign relation $R_{\phi^-} = -(\phi^- * \phi^-)_{\text{math}}$ was initially handled incorrectly), not a post-hoc fit of the coefficient to the data. The rigorous derivation closes the convention. The derivation also yields two equivalent global conventions: the Connes-operator convention used throughout the atlas (coefficient -2) and the Weil-Q convention (coefficient $+2$), which differ by an overall sign on the prime part. The atlas does not depend on which convention is chosen, as long as it is applied consistently; all numerical signs are consistent with the Connes-operator convention.

IX Outlook

The atlas delivers a precise diagnosis: predictive closure of $C^{(2)}_\chi$ via Galerkin numerics is structurally impossible, and the way forward is an analytic control of the ground-state difference that bypasses Galerkin entirely. Six concrete next steps are indicated.

1. **Route D: χ-twisted CCM transfer.** The first post-Zookeeper task is to construct the Dirichlet analogue of the CCM/Zookeeper operator: a χ-twisted Epstein–Mellin transform, a conductor rank-one term, and a χ-weighted arithmetic perturbation. The atlas quantities G_χ, R_χ, and $\eta^{\pm}_\chi$ should then be re-read as diagnostics for the corresponding defect norm, rather than as direct Paley-Wiener predictors.

2. **Paley-Wiener asymptotic analysis of $\phi^{\pm}_\chi$.** For the trivial character the asymptotic ground state ϕ_∞ is known explicitly from the Paley-Wiener machinery of [4] (§5–§6) under the REG-λ regularity assumption. The adaptation to Dirichlet L-functions amounts to identifying the χ-twisted analogue of the relevant Fourier-multiplier bound and the rational-function contour to which the residue-theorem is applied. Three of the five necessary steps transfer mechanically (cf. [8], Thm. 3.1); the remaining two require a character-specific treatment of $L'/L(s, \chi)$ in the critical strip. This is the single highest-leverage technical task.

3. **Branch-local Hilbert-Pólya framework ("NE-B Boundary").** Building on [4] (§4 NE-A, §5 NE-B) and on the Selberg analogue [13], the next major project in the meta-programme is a structural theorem identifying *which branches of the zeta-function tree admit a Hilbert-Pólya interpretation, and which do not.* The present atlas contributes the first quantitative evidence that low-conductor Dirichlet characters populate the second class (no branch-local Hilbert-Pólya structure at finite λ), sharpening the boundary theorem that will organize the tree.

4. **Conductor extension $D \leq 100$.** Ten characters is a small atlas. Extending to all primitive real characters with $D \leq 100$ (≈ 30 additional characters including a few complex conjugates grouped in paired spaces) would enable statistical tests of the parity/root-number/archimedean classification against a distribution of observed gap signs. The Galerkin cost is $\mathcal{O}(D \cdot N^3)$; at $N = 600$ this is ~ 10 hours of CCX13 compute per ten characters, a one-weekend job.

5. **Paired-space formalism for complex characters.** Complex Dirichlet characters come in conjugate pairs $\chi \oplus \bar{\chi}$; the Weil quadratic form on $L^2[-L, L] \otimes \mathbb{C}^2$ is hermitian but not self-adjoint. The analytic framework of §II generalizes, but the sector kernel (Theorem II.2) acquires a Gauss-sum

phase ([6], Prop. 5.2). A clean exposition is a prerequisite for testing the root-number axis (§B) against non-trivial values $W \in \{+1, -1, \pm i\}$.

6. **Theoretical derivation of the Siegel-Walfisz compression.** The empirical dichotomy of Remark IV.1—between characters with stable $\mathrm{gap}_{\mathrm{gal}}^{(N=200)}$ and those that are compressed by a factor ~ 0.35 at $N = 600$—asks for a theoretical classification in terms of the character-specific Siegel-Walfisz constant. A candidate formulation is that χ with a genuine low-prime bias (e.g. χ_{33}, χ_{13}) has a prime sum that saturates the Siegel-Walfisz upper bound in the relevant window and is therefore N-stable, whereas χ with a balanced prime sum (e.g. χ_{12}) has a prime sum below the upper bound and is therefore compressible.

A Computational details

A Galerkin basis and matrix assembly

We use the standard Fourier orthonormal basis of $H_L^{\pm}$: $\{e_0^+ = (2L)^{-1/2}\} \cup \{e_n^+(x) = L^{-1/2}\cos(n\pi x/L)\}_{n\geq 1}$ for the even sector and $\{e_n^-(x) = L^{-1/2}\sin(n\pi x/L)\}_{n\geq 1}$ for the odd sector. The Galerkin-truncated operators $W_{\chi,N}^{\pm}$ at basis size N are assembled by the function `build_W`$(\chi, \lambda, \mathrm{sector}, N)$ of the script `arch_term_analysis.py`, which decomposes $W^{\pm} = W_{\mathrm{arch}}^{\pm} + W_{\mathrm{prime}}^{\pm}$ where

(i) $W_{\mathrm{arch}}^{\pm}$ is obtained from the Fourier image of $\rho_\chi(t) = \log(q/\pi) + \Re\,\psi(1/4 + it/2)$ on a grid of spacing $\Delta t = 10^{-2}$, truncated at $|t| \leq T_{\mathrm{arch}} = 30$ (chosen so that the Gaussian ground-state spectrum is contained within numerical accuracy).

(ii) $W_{\mathrm{prime}}^{\pm}$ is obtained by summing $\chi(p)^m \log p / p^{m/2}$ over all (p, m) with $p \nmid q$ and $m \log p \leq 2L$, evaluating the basis-function overlaps analytically (for cosine-cosine) or by Simpson integration (for sine-sine off-diagonal terms).

The smallest eigenvalue and its eigenvector are extracted by Lanczos inverse iteration from `scipy.sparse.linalg.eigsh` with tolerance 10^{-10} and `sigma`$= -\log\lambda$ as a spectral shift.

B Autocorrelation evaluation

The spatial-domain autocorrelation $(\phi^{\pm} * \phi^{\pm})(t)$ is computed on a uniform grid of $n_{\mathrm{grid}} = 4096$ points on $[-2L, 2L]$ via `numpy.correlate` after zero-padding, then interpolated at the prime abscissae $t = m \log p$ by cubic `scipy.interpolate.interp1d`. The resulting realization of S_χ, called S_{exact} in our scripts, differs from the matrix-based realization $\mathrm{prime\_diff}_{\mathrm{matrix}}$ by floating-point and interpolation errors of order 10^{-3} (1–2% of typical $|S|$-values); this is the source of the $N = 600$ sign flips at χ_{29} and χ_{60} documented in §IV.

C Scripts

All data in this paper were generated by `arch_term_analysis.py` (for $N = 200$ local), `server_n600_full_analysis.py` (for $N = 600$ server run, 1.95 h on Hetzner CCX13), and `chi21_n_convergence.py` (for Table 5). Post-processing is done by `ground_state_difference_analysis.py`. JSON outputs are in `CORE/zoo-mapping/_results/`; Markdown summaries in the same directory.

B Sign-convention status

The coefficient -2 in (13) is rigorously derived from the Weil explicit formula in [7], Theorem 5.1, by tracking all six relevant sign sources (Weil-formula prime-side minus, operator convention, gap orientation, parity action on basis, convolution-vs-autocorrelation parity identity, and the real-character absorption factor). The derivation yields *two equivalent conventions*:

1. the **Connes-operator convention** $\tilde{A}_\chi$, in which the prime terms are added with positive sign in the matrix assembly; this matches the script `build_W` (i.e. $\kappa_\chi^{\mathrm{script}}(u) = +\sum w_{p,m}[\delta(u - m\log p) + \delta(u + m\log p)]$) and yields coefficient -2;

2. the **Weil-Q convention** Q_χ used in (5), in which κ_χ carries a leading minus sign; here the analogous derivation yields coefficient $+2$ (as in [6], Cor. 4.4).

The two operators satisfy $\tilde{A}_\chi^{\mathrm{prim}} = -Q_\chi^{\mathrm{prim}}$ (the prime parts differ by an overall sign); both are valid descriptions of the same spectral problem, but they give oppositely signed gaps. The atlas and the script consistently use the Connes-operator convention throughout, so the coefficient -2 in (13) is the correct one.

A second subtlety concerns the object being summed. The numerical implementation of (13) (R1, spatial autocorrelation) uses `np.correlate`, which computes the *autocorrelation* $R_{\phi^\pm}(u) := \int \phi^{\pm}(s)\phi^{\pm}(s-u)\,ds$. For the *even* ground state ϕ^+ one has $R_{\phi^+} = (\phi^+ * \phi^+)_{\mathrm{math}}$, but for the *odd* ground state ϕ^- the parity identity gives $R_{\phi^-} = -(\phi^- * \phi^-)_{\mathrm{math}}$ (proof: substitution $v = -s$ in the integral $\int \phi^-(s)\phi^-(s-u)\,ds$, using $\phi^-(-v) = -\phi^-(v)$). Consequently, the numerical expression $\Delta_\chi^{\mathrm{num}} := R_{\phi^+} - R_{\phi^-}$ computed by the script equals $(\phi^+ * \phi^+)_{\mathrm{math}} + (\phi^- * \phi^-)_{\mathrm{math}}$ in mathematical convolutions—a *sum*, which is precisely the right-hand side of (10). Thus Δ_χ as defined in (10) agrees identically with the script-computed quantity $\Delta_\chi^{\mathrm{num}}$. The coefficient -2 in (13) is then the unique value that makes the gap identity (Theorem III.2) hold. See [7], §4 and Theorem 5.1, for the full derivation. All reported signs and all tables (Tables 2, 1, 4, 5) are internally consistent and have been verified against a direct Galerkin diagonalization.

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